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Research on the Mechanism of the Multimodal Sustained Usage of Sport Drones from the Perspective of the Low-Altitude Economy

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Abstract

Against the backdrop of the low-altitude economy, with the widespread application of drones in sports scenarios, the driving mechanism of users' long-term usage intention has become a key issue in technology adoption research. To investigate the critical factors influencing the continuous use of drone products by sports-involved populations, this study builds a factor model for users' continuous use of drones. It is based on the Unified Theory of Acceptance and Use of Technology (UTAUT) model, integrating the UTAUT2 model and specific user needs in sports scenarios. Both traditional structural equation modeling (SEM) and Bayesian structural equation modeling (BSEM) are employed for empirical testing. Through the analysis of 297 valid questionnaire responses, it is found that the Bayesian approach yields a better fit. Performance Expectancy, Effort Expectancy, Social Influence, Facilitating Conditions, Hedonic Motivation, Safety and Environmental Compatibility all exert significant positive impacts on users' continuous usage intention, with Effort Expectancy having the most prominent influence. On this basis, service strategies for drone brands are proposed to support product design and service provision. This study preliminarily indicates that Bayesian analysis possesses advantages and potential in this field. Meanwhile, the factor model for users' long-term drone usage can meet the development needs in sports scenarios, and it has strong feasibility as a design model for users' long-term drone usage.

Keywords: sport consumer drones; extended UTAUT theory; Bayesian SEM; mediating effect model; low-altitude economy

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1. Introduction

In recent years, with the advancement of smart city construction and the continuous development of the global economy, the emerging industry of the low-altitude economy has gradually attracted widespread attention and is increasingly becoming a key engine driving economic growth and industrial innovation. The low-altitude economy is a comprehensive economic form centered on the construction of the entire industrial chain, driven by various low-altitude flight activities of manned and unmanned aerial vehicles, and driving the integrated development of related fields through radiation [1]. Consumer drones, with their advantages such as being light weight, low cost, and environmentally friendly, have a broad market in fields such as logistics and transportation, agricultural and forestry irrigation,

urban transportation, safety and rescue, and entertainment and leisure, injecting new vitality into the low-altitude economy [2–5]. According to statistical data, the compound annual growth rate (CAGR) of China's drone industry is approximately 13.8%. There are over 15,000 drone enterprises nationwide, with more than 700,000 registered users of drone owners and over 1 million registered drones. It is estimated that by 2026, the global drone market will reach 40.7 billion USD.

With the deep integration of intelligent technologies and sports, consumer-grade drones have now been applied in sports-related scenarios such as training, competitions, and fitness. Gubdes Russomanno et al. mounted cameras on drones to capture videos of sports training scenarios (e.g., tennis, football, and ultimate frisbee) from a bird's-eye view, which are used to analyze players' positions and trajectories [6]. Jacobsson et al. proposed the use of micro-drones equipped with depth cameras, Cubemos Skeleton tracking and Nuitrack Pro for real-time motion capture and analysis in outdoor sports training scenarios [7]. Balasubramaniam et al. applied drones to track and field pacing services, exploring the potential of drones as pacers and video recorders in individual running training [8].

In live broadcasts of sports events, drones have become indispensable shooting tools, providing audiences with new viewing perspectives, thereby enhancing the immersion and viewership experience of the events [9]. Drones can also interact with on-site audiences through sound and light signals, conveying cheers and encouragement to athletes [6]; in addition, they provide real-time data support for refereeing decisions and sports performance analysis [6].

In terms of daily sports and fitness, Chaitika et al. applied drones to a home fitness guidance system for elderly users, using micro-drones to demonstrate exercise movements and assisting the elderly in home workouts through voice feedback [10]. In the research by Christian Eichhorn et al., drones were extended to an interactive product used in multi-player sports games [11].

Although consumer-grade drones have demonstrated numerous innovative advantages in sports scenarios, they still face multiple challenges in the process of practical promotion. Existing studies tend to focus more on exploring technologies for improving drone performance and strategies for optimizing application paths, while paying less attention to the role played by consumers (i.e., the sports population) in drone consumption. As the core service targets of consumer-grade drones, the sustained usage intention of the sports population is crucial to the market promotion and sustainable operation of products [12]. Therefore, it is necessary to conduct more detailed research from the perspective of the sports population to further explore how to enhance their intention to continuously use drones.

Sustained usage intention reflects users' long-term acceptance of a product after initial use. According to previous studies, public acceptance and the sustained use of drones are comprehensively influenced by factors such as the functional practicality of drones, operational convenience, the degree of public awareness and understanding of drones, and the convenience of drone-related supporting service systems [13]. Therefore, this study intends to explore long-term design strategies for drones targeting the sports population based on the framework of the Unified Theory of Acceptance and Use of Technology (UTAUT). This research aims to identify key factors affecting the long-term usage intention of the sports population, optimize drone product services for sports scenarios, and thereby enhance user experience and user stickiness, fostering consumer demand for drones in the sports field, providing theoretical support and practical guidance for related industrial research, and contributing to the development of the low-altitude economy.

The remainder of this paper is organized as follows: In Section 2, we review the background and research development of the UTAUT theoretical model, as well as user studies on drones, thereby deriving the hypothetical model. Section 3 describes the data collection and data analysis methods. Section 4 presents the results of the data analysis. Section 5 discusses the test results and draws conclusions; it also clarifies the academic and practical implications, limitations, and future research directions.

2. Theoretical Background and Hypotheses

2.1. Theoretical Background

The Unified Theory of Acceptance and Use of Technology (UTAUT) was proposed by Venkatesh [12], aiming to integrate various previous technology acceptance models and explore the key factors influencing technology usage intention and behavior. The UTAUT theory has not only been widely validated in the field of information technology but also proven to have high predictive power and effectiveness in studies on the acceptance of various emerging technologies. Compared with other technology acceptance models, a prominent advantage of the UTAUT model is that it can comprehensively and systematically explain users' acceptance and usage intention of technology in different environments by integrating the core variables of multiple theories [12].

Venkatesh and other scholars reviewed and compared eight classic models: the Theory of Reasoned Action (TRA), Technology Acceptance Model (TAM), Motivational Model (MM), Theory of Planned Behavior (TPB), TAM-TPB integrated model, Model of PC Utilization (MPCU), Diffusion of Innovations Theory (IDT), and Social Cognitive Theory (SCT). They extracted common influencing factors and constructed a unified framework, namely the UTAUT model, to enhance the accuracy and universality of the explanatory power regarding users' technology acceptance and usage behavior [12]. The four main constructs of the UTAUT model are Performance Expectancy, Effort Expectancy, Social Influence, and Facilitating Conditions, along with four moderating variables: age, gender, experience, and voluntariness. Ultimately, the model takes "Behavioral Intention (BI)" as the outcome variable. These constructs not only cover users' cognitive evaluation of technology but also take into account the influences of social and technological environments [14].

Advancements in technology and the iteration and expansion of technological products prompted Venkatesh et al. to extend the UTAUT model. In 2012, based on the original model, Venkatesh et al. added three new variables, Hedonic Motivation, price value, and habit, resulting in the UTAUT2 model. The UTAUT2 model is particularly applicable to studies in commercial scenarios and concerning users' sustained usage intention [15]. The newly added dimensions addressed the limitation of the original model, which primarily focused on evaluating external motivation while having a limited evaluation of internal motivation [16].

The UTAUT theoretical model provides in-depth insights into users' acceptance of technologies and services from multiple dimensions, and exhibits strong adaptability in studying the factors influencing the sports population's use of drones. As a technology-empowered product, drones allow for the extraction of factors influencing users' usage behavior based on the UTAUT theoretical model. Currently, the UTAUT theoretical model has been adopted by many scholars in the field of drone products and related services, being applied to studies on users' usage behavior and sustained usage intention in areas such as logistics distribution services [17] and medical transportation services [18]. Therefore, this study intends to incorporate the indicators of the UTAUT into the factors influencing the sports population's usage intention of drone products, propose an extended UTAUT model tailored to the research objectives, and analyze "Continuous Intention (CI)" and "Actual Usage Behavior (UB)" as key indicators.

2.2. Hypothetical Model

The hypothetical model in this study is based on the UTAUT model. Considering that technology acceptance in sports scenarios relies more on the compatibility of product functions and scenarios rather than demographic characteristics such as age and gender, and these factors, which act as moderating variables, have no significant explanatory power for usage, they are excluded; meanwhile, since drones are emerging products in the sports field and users have not formed stable usage habits, the habit variable is also excluded. After excluding moderating factors such as age, gender, experience, and voluntariness from the original model that are not applicable to this study, the research supplements and incorporates some variables from the UTAUT2 model. As well as the core elements of drone design in the sports field, they are used as independent variables of the research to increase the predictive ability of the model, forming a hypothetical model for studying the factors of long-term drone usage among the sports population. The independent variables of the hypothetical model are Performance Expectation, Effort Expectation, Social Influence, Facilitating Conditions, Hedonic Motivation, Safety, and Environmental Compatibility, and the dependent variables of the research are Continuous Usage Intention and Actual Usage Behavior. The constructed model is shown in Figure 1.

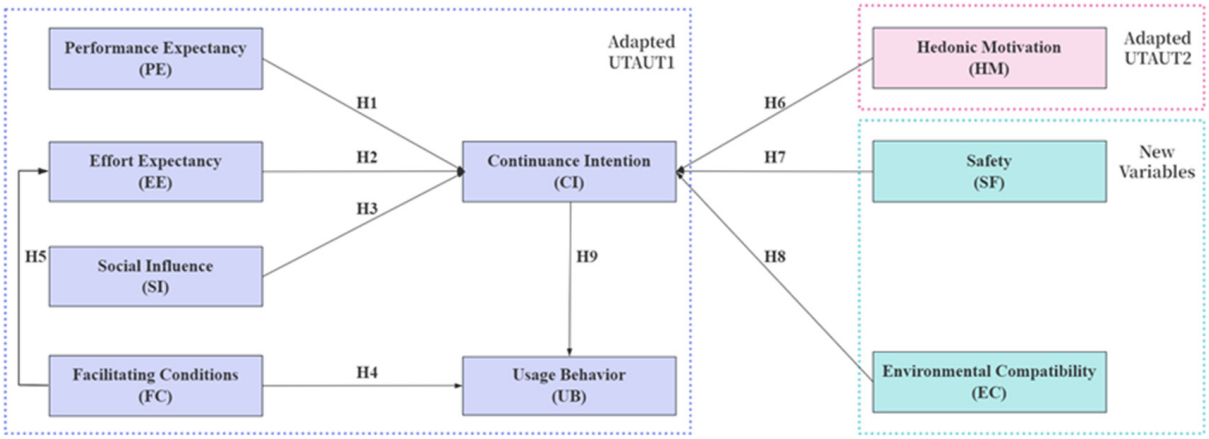


Figure 1. Hypothetical model (drawn by the author).

In the UTAUT model, Performance Expectancy is defined as the degree to which an individual believes that using the system will help him or her achieve improved job performance. This construct is derived from five constructs: Perceived Usefulness (TAM/TAM2 and C-TAM-TPB), Extrinsic Motivation (MM), Job Fitness (MPCU), Relative Advantage (IDT), and Outcome Expectation (ST) [12]. In this study, for drone products targeting users in the sports population, Performance Expectancy refers to the practicality of drone functions, which is mainly manifested in the level to which the functions carried by the drone product improve users' fitness exercise effects and enhance users' exercise efficiency. Therefore, the following hypothesis is proposed:

H1. *Performance Expectancy has a significant positive impact on users' continuous use intention.*

Effort expectation is analogous to the perceived ease of use in the Technology Acceptance Model (TAM), which is related to the convenience of using information systems and positively correlated with user satisfaction [12]. As an emerging technology in the sports field, drones have a certain operational threshold, so effort expectation mainly refers to the ease with which users can operate the drone system. Sports populations typically focus more on how to enhance sports performance and experience through technology rather than investing excessive energy in learning complex operating systems [19]. Intuitive

interface design, intelligent automatic flight modes, simple one-click control functions, etc., can greatly reduce the operational difficulty for athletes, helping sports populations focus on the sport itself rather than being troubled by complex technical operations. Based on this, the following hypothesis is proposed:

H2. *Effort expectation has a significant positive impact on users' continuous use intention.*

Social Influence pertains to the impact of others' opinions on an individual's technology adoption. According to UTAUT theory, Social Influence directly affects continuous use intention. For drone technology, Social Influence is associated with public awareness and attitudes toward drones. Venkatesh [12] defined SI as three elements: image, social factors, and subjective norms. Individuals are more likely to adopt and use a piece of technology when they perceive approval and support from others [12]. Research by Aydin et al. indicates that the public acceptance of drone products is generally influenced by societal perceptions [13]. Moreover, sports populations often exhibit strong group identity and social interaction needs [6], suggesting that their technology use is susceptible to peer influence. Based on this, the following hypothesis is proposed:

H3. *Social Influence has a significant positive effect on users' continuous use intention.*

Facilitating Conditions refer to the availability of resources and support required for the effective use of the technology. Specifically, it is defined as the degree to which an individual believes that organizational and technical infrastructure exists to support the use of information systems (IS) [12]. It is regarded as a belief related to an individual's control over the use of information systems. Facilitating Conditions do not affect users' continuous use intention but directly impact users' actual use behavior. In this study, Facilitating Conditions mainly measure the supporting services and technical support of drone products. Meanwhile, according to Venkatesh et al., Facilitating Conditions can promote effort expectation to a certain extent [15]. In the context of drones, this promotion is reflected in that the supporting services of drone products provide assistance to sports populations when they encounter operational difficulties, thereby reducing users' operational learning costs. Based on this, the following hypotheses are proposed:

H4. *Facilitating Conditions have a significant positive impact on users' continuous use behavior.*

H5. *Facilitating Conditions have a significant positive impact on effort expectation.*

In the UTAUT2 theoretical model, Venkatesh et al. introduced Hedonic Motivation as a key construct supplementary to Performance Expectancy, thereby expanding the context of consumer technology use from pure instrumentality to simultaneously encompassing entertainment and enjoyment. It specifically refers to the feelings of pleasure, amusement, or fun that users derive when using a certain technology or product. This motivation emphasizes the entertainment value of technology and the pleasure during use, rather than mere functionality or efficiency [15]. Considering the characteristics of sports populations who pursue challenging, fun, and personalized experiences [19], drones not only serve as tools to provide sports data and shooting support but also enhance the fun and attractiveness of the sports process, becoming their entertainment partners in sports. Therefore, Hedonic Motivation is also incorporated into the model construction in this paper, and the following hypothesis is proposed:

H6. *Hedonic Motivation has a significant positive impact on users' continuous use intention.*

In addition, this study identifies the main influencing factors of consumers' use of drone technology in sports scenarios by reviewing existing research, and incorporates the Safety and Environmental Compatibility of drone technology into the evaluation framework.

Kim et al.'s research has shown that the safety of drone use is one of the important factors affecting users' interest in drone functions. The good safety performance of drone products can reduce users' concerns about drone accidents, and improving drone safety can enhance people's experience and utilization rate of drones [20]. Therefore, in this study, the Safety of drones is defined as the ability of drone products to ensure that they will not cause potential harm or loss to humans, property, the environment, and air traffic systems during use, operation, and maintenance. Based on this, the following hypothesis is proposed:

H7. *Safety has a significant positive impact on users' continuous use intention.*

The Environmental Compatibility of drone consumption refers to the ability of drones to operate stably under various environmental conditions, such as different weather patterns, geographical terrains, and communication settings. The design of consumer products targeting sports enthusiasts must account for diverse performance environments to meet users' evolving needs across different athletic scenarios. Consequently, Environmental Compatibility emerges as a critical determinant influencing users' adoption of sports-related products [20]. To satisfy the functional requirements of sports populations, drones must seamlessly integrate into users' complex activity environments during actual use. Based on this premise, the following hypothesis is proposed:

H8. *Environmental Compatibility has a significant positive effect on continuous use intention.*

In the UTAUT2 model proposed by Venkatesh [15], user's usage intention refers to the individual's subjective possibility or degree of intention to use a certain technology in the future. User's actual behavior refers to the frequency and breadth of the respondent's use of the technology in a real environment, which is the real usage performance after intention. In the original UTAUT model, usage intention is regarded as a direct antecedent factor affecting actual behavior, and the relationship between the two is also moderated by "usage experience" in the consumer context: the more experience, the weaker the influence of intention on behavior [15]. Therefore, the drone usage intention of sports participants is a direct antecedent factor driving their subsequent actual use. As a new type of consumer product, drone consumption has not been widely used in the sports field, and the sports population is not familiar with it, so it is less affected by experience factors. Therefore, the hypothesis is proposed:

H9. *Users' Continuous Usage Intention has a significant positive impact on Actual Usage Behavior.*

3. Research Methods

3.1. Sampling Procedure and Data Collection

This study will use quantitative analysis to test the hypothetical model of the long-term use of drone products by sports populations in sports scenarios. Therefore, based on the demand characteristics of the sports population and the product usage environment, we modified several existing valid questionnaires to create a new questionnaire. Each construct was selected with 3 questions, which were either derived from the original UTAUT model or from papers that have tested the proposed factors. The questions were measured using a 5-point Likert scale ranging from strongly disagree to strongly agree, and the questions are presented in Table 1. In addition, the questionnaire also included surveys on age, gender,

occupation, daily exercise frequency, and drone usage experience. To ensure the accuracy and reliability of the questionnaire, before distribution, it was carefully reviewed by experts in the field, and appropriate improvements and adjustments were made to the minor issues of the questionnaire accordingly.

Table 1. Constructs and items used for online survey.

Construct	Item		Source Adapted
Performance Expectancy (PE)	PE1	Using drones can help improve my sports performance.	Venkatesh et al. (2012) [15]; Venkatesh et al. (2003) [12];
	PE2	Using drones significantly enhances my sports performance.	
	PE3	The exercise evaluation function of drones is very important to me.	
Effort Expectancy (EE)	EE1	I think it will be easy for me to learn how to use drone.	Venkatesh et al. (2012) [15]; Schmidt et al. (2024) [21];
	EE2	I think my interaction with the service via the mobile application will be clear and understandable.	
	EE3	I think it will be easy for me to become skillful at using drone.	
Social Influence (SI)	SI1	People who are important to me will think that I should use drone.	Venkatesh et al. (2003) [12]; Zhou et al. (2020) [22];
	SI2	People who are important to me will think that I should keep using drones during workouts.	
	SI3	People whose opinion I value will prefer that I use drone.	
Facilitating Conditions (FC)	FC1	I think I will have the resources necessary to use drone.	Venkatesh et al. (2012) [15]; Venkatesh et al. (2003) [12]; Eißfeldt et al. (2020) [23];
	FC2	I think I will get help from the service provider if I have any difficulty using the drone.	
	FC3	While using drones, any technical issues can be conveniently resolved.	
Hedonic Motivation (HM)	HM1	Using drone during workouts will be fun.	Schmidt et al. (2024) [21]; Venkatesh et al. (2012) [15];
	HM2	Using drone during workouts will be enjoyable.	
	HM3	Using drone during workouts will be enjoyable entertaining.	
Safety (SF)	SF1	Drone products will ensure no harm to personnel or property during operation.	Smith et al. (2022) [24]; Gupta et al. (2021) [25];
	SF2	Drone products can better protect users and reduce injury risks.	
	SF3	During unexpected situations, drone products can provide emergency assistance.	
Environmental Compatibility (EC)	EC1	Drones can operate effectively across diverse environmental conditions.	Fehling et al.(2023) [26]; Shankar et al.(2024) [27]; Wang et al.(2023) [28]
	EC2	Drones maintain user experience quality across diverse environments.	
	EC3	I believe drones can adapt to diverse environments to fulfill my needs.	
Continuance Intention (CI)	CI1	I plan to use drone frequently when available in the future.	Venkatesh et al. (2012) [15]; Venkatesh et al. (2003) [12];
	CI2	I will continue using drone products regularly in the future.	
	CI3	I would recommend drone products to other workout lovers.	
Usage Behavior (UB)	UB1	I will regularly use drone products during workouts.	Venkatesh et al. (2012) [15]; Venkatesh et al. (2003) [12];
	UB2	I will use drone products to interact during exercise.	
	UB3	I will reuse drone products during workouts and maintain ongoing usage.	

To enhance collection efficiency, we primarily used online distribution. For data collection, convenience sampling was employed to distribute questionnaires to diverse groups across China, aligning with the research focus on drone usage in sports scenarios. The official survey was conducted from 1 December 2024 to 15 January 2025. To cover target populations including athletes and sports practitioners, respondents were selected via a combination of random sampling and targeted scenario outreach: potential participants were first identified through random sampling, followed by the targeted distribution of questionnaire links to college sports enthusiasts and sports communities via online channels to improve efficiency. A total of 304 responses were collected, with 297 valid after validity checks (excluding incomplete or logically inconsistent submissions).

Table 2 presents the characteristics of the respondents. The data show that among the respondents, the proportion of female respondents (47.1%) was slightly lower than that

of male respondents (52.9%). Additionally, 73.4% of the respondents were aged between 18 and 24. The occupational distribution of the respondents was mainly concentrated in students (52.9%). Overall, the distribution of respondents' weekly exercise frequency was relatively balanced, and most respondents had the habit of exercising. The number of weekly exercise times for most users was concentrated in 3–4 times (103 people, 34.7%) and 1–2 times (84 people, 28.3%); 59 people (19.9%) exercised 5–6 times a week, and 46 people (15.5%) exercised 7 times or more; only 5 people (1.7%) exercised less than once a week, accounting for a very small proportion.

Table 2. Characteristics of respondents (n = 297).

Category	Items	Frequency	Percentage (%)
Gender	Male	157	52.9%
	Female	140	47.1%
Age	18–24	218	73.4%
	25–34	31	10.4%
	35–44	21	7.1%
	45–54	16	5.4%
	>55	11	3.7%
Careers	Student	157	52.9%
	Self-employed	34	11.4%
	Corporate Employee	63	21.2%
	Public Employee	36	12.1%
	Other	7	2.4%
Exercise frequency per week	0	5	1.7%
	1–2	84	28.3%
	3–4	103	34.7%
	5–6	59	19.9%
	>7	46	15.5%
Usage period of drone	<1 month	51	17.2%
	1–3 months	45	15.2%
	3–6 months	97	32.7%
	6–12 months	42	14.1%
	>1 year	62	20.9%
Primary drone applications	Photography	150	50.5%
	Videography	149	50.2%
	Recreational Use	143	48.1%
	Sports Analytics	91	30.6%
	Other	5	1.7%

In terms of drone familiarity, nearly one-third of users have used drones for 3–6 months (97 people, 32.7%); users with more than one year of experience account for 20.9% (62 people); novice users with less than 1 month and 1–3 months of use account for 17.2% (51 people) and 15.2% (45 people), respectively; and 42 people (14.1%) have used drones for 6–12 months. Overall, medium-term users (with 3–12 months of use) account for a relatively high proportion. In terms of drone usage, photography and video recording are the two main purposes, with 150 people (50.5%) and 149 people (50.2%) choosing them, respectively; 143 people (48.1%) use drones for companionship and entertainment; 91 people (30.6%) use them for sports analysis; and only 5 people (1.7%) use them for other purposes. It can be seen that image acquisition remains the primary motivation for users to choose drones.

3.2. Variable Measurement

The research model comprises nine constructs: Performance Expectancy (PE), Effort Expectancy (EE), Social Influence (SI), Facilitating Conditions (FC), Hedonic Motivation (HM), Security (SF), Environmental Compatibility (EC), Continuance Intention (CI), and Actual Usage Behavior (UB). Specifically, PE reflects users' perception of the practicality of drones; EE embodies users' subjective judgment on the operational simplicity of drones; SI represents the influence of social environment on technology adoption; FC refers to users' accessibility to necessary resources, services, and technical support during actual drone operation; HM denotes the fun, entertainment, and pleasure experienced by users

when using drones in sports; SF refers to users' subjective perception of whether drones are safe and whether potential risks exist during flight and interaction; EC refers to the ability of drones to operate stably in various sports scenarios (such as different climates, terrains, and lighting conditions); CI denotes users' intention and tendency to continuously use drone products in sports scenarios in the future; and UB refers to users' actual usage frequency and behavioral performance of drones in real sports scenarios, serving as the final validation of Continuance Intention.

In the hypothesis model testing, this study mainly uses structural equation modeling (SEM) to evaluate and analyze the hypothetical model. SEM is a widely accepted and frequently used research hypothesis testing tool in psychology and social sciences, which is used to test the direct and indirect effects between latent variables, so as to test and revise theoretical hypotheses [29]. It has been confirmed as a powerful multivariate tool for testing, confirming, or comparing theories [30]. This study will use Mplus to measure the model through the traditional maximum likelihood estimation method and conduct comparative analysis through Bayesian SEM with the R language. The traditional SEM estimates model parameters through the most commonly used maximum likelihood (ML) method, but it is only approximately valid when the sample size is very large; when the sample size is small, the estimation accuracy and fit test seriously deviate from the theory [31]. In this study, the sample size is 297. Although it meets the minimum requirements for traditional SEM analysis, the model contains multiple latent variables and path relationships, and the number of parameters is large. The traditional maximum likelihood estimation is prone to estimation bias with small and medium samples, or unstable standard errors and unreliable marginal significance. The Bayesian method does not rely on asymptotic theory. It directly infers parameters by constructing posterior distributions and can still obtain more accurate estimates with small samples. It provides more and more precise information on estimated model parameters (i.e., information about the joint probability distribution of parameters) and it incorporates prior information, which can specify informative priors for parameters that are difficult to estimate or rich in prior knowledge, making the estimation more accurate and interpretable [29].

In the research on drone Continuance Intention, the commonly used method is the traditional structural equation modeling (SEM), while the application of Bayesian estimation methods is rare. Therefore, this study adopts a comparative approach between traditional SEM and Bayesian SEM, and determines which method is more suitable for the hypothetical model of this study by comparing the data results of the two.

4. Analysis and Results

4.1. Normality Test

To ensure the reliability of the statistical basis for the comparative analysis between traditional structural equation modeling (SEM) and Bayesian structural equation modeling (BSEM), this study first tested the normality of the data. The results in Table 3 show that the means of all observed variables ranged from 3.047 to 3.687, with standard deviations distributed between 1.435 and 1.929. The absolute values of skewness ranged from 0.034 (CI2) to 0.671 (EC3), and the absolute values of kurtosis ranged from 0.985 (EC3) to 1.921 (UB2). According to Kline (2015)'s multivariate normality assessment criteria (absolute skewness < 3, absolute kurtosis < 10) [32], the skewness ($|\text{Skewness}| \leq 0.671$), and kurtosis ($|\text{Kurtosis}| \leq 1.921$) of all measurement indices in this study were significantly lower than the critical thresholds. The data distribution exhibited characteristics of slight left skewness (negative skewness) and platykurtosis (negative kurtosis), but overall met the multivariate normality assumption, supporting the applicability of subsequent parameter

estimation methods for SEM and BSEM, and ensuring the statistical robustness of the measurement model (as shown in Table 3).

Table 3. Normality of data distribution (n = 297).

Construct	Item	Mean	Std. Dev	Skewness	Kurtosis
PE	PE1	3.535	1.440	−0.480	−1.167
	PE2	3.138	1.521	−0.102	−1.453
	PE3	3.158	1.522	−0.131	−1.443
EE	EE1	3.296	1.657	−0.326	−1.556
	EE2	3.448	1.667	−0.476	−1.464
	EE3	3.047	1.645	−0.035	−1.631
SI	SI1	3.562	1.501	−0.532	−1.205
	SI2	3.569	1.541	−0.536	−1.389
	SI3	3.367	1.541	−0.357	−1.389
FC	FC1	3.441	1.499	−0.411	−1.270
	FC2	3.098	1.538	−0.103	−1.467
	FC3	3.114	1.457	−0.134	−1.300
HM	HM1	3.138	1.556	−0.123	−1.513
	HM2	3.451	1.495	−0.417	−1.309
	HM3	3.367	1.523	−0.404	−1.305
SF	SF1	3.512	1.522	−0.512	−1.228
	SF2	3.327	1.508	−0.288	−1.363
	SF3	3.428	1.607	−0.412	−1.449
EC	EC1	3.606	1.519	−0.580	−1.200
	EC2	3.306	1.552	−0.291	−1.445
	EC3	3.687	1.435	−0.671	−0.985
CI	CI1	3.172	1.898	−0.170	−1.880
	CI2	3.034	1.903	−0.034	−1.913
	CI3	3.135	1.900	−0.134	−1.896
UB	UB1	3.141	1.916	−0.139	−1.908
	UB2	3.128	1.929	−0.136	−1.921
	UB3	3.141	1.894	−0.150	−1.881

4.2. Evaluation of C-SEM Model

4.2.1. Evaluation of Measurement Model CFA

This study comprehensively evaluates the reliability and validity of the measurement model through standardized factor loadings, squared multiple correlations (SMCs), and average variance extracted (AVE). The standardized factor loadings of all items significantly exceed the minimum standard of 0.50 (actual range: 0.774–0.984) [33], among which the Performance Expectancy (PE) item PE1 reaches 0.880 and the Actual Usage Behavior (UB) item UB3 reaches as high as 0.984. The squared multiple correlations (SMCs) reflect the explanatory power of each dimension for the items, with all items' SMC values exceeding the ideal level of 0.50 (actual range: 0.575–0.968). As an indicator of internal consistency, the composite reliability (CR) of all constructs exceeds the standard requirement of 0.7 (actual range: 0.832–0.984), indicating that the measurement model has excellent reliability. The average variance extracted (AVE), used to measure the average explanatory power of dimensions for items, shows that all constructs' AVE values are higher than the ideal standard of 0.50 (actual range: 0.623–0.954), confirming that the convergent validity of all variables meets the requirements of the measurement model. Overall, the data show that all measurement indices meet or significantly exceed the standard thresholds, and the model has excellent reliability and validity levels (as shown in Table 4).

Discriminant validity assesses whether the items measuring research variables differ from those measuring other variables, emphasizing their uniqueness. In essence, it evaluates whether items measure distinct constructs. Discriminant validity is strong if the square root of a variable's AVE exceeds its correlations with other constructs [32]. In this study, the analysis shows that the square root of the AVE for each variable is greater than its Pearson's correlation coefficients, as presented in the rows and columns of Table 5.

Table 4. Results of validity and reliability analysis (n = 297).

Construct	Item	Factor Loading	p-Value	SMC	C_R	AVE
PE	PE1	0.880	***	0.774	0.866	0.683
	PE2	0.805	***	0.648		
	PE3	0.791	***	0.626		
EE	EE1	0.918	***	0.843	0.922	0.797
	EE2	0.867	***	0.752		
	EE3	0.892	***	0.796		
SI	SI1	0.799	***	0.638	0.864	0.679
	SI2	0.860	***	0.740		
	SI3	0.812	***	0.659		
FC	FC1	0.775	***	0.601	0.855	0.663
	FC2	0.819	***	0.671		
	FC3	0.848	***	0.719		
HM	HM1	0.788	***	0.621	0.857	0.667
	HM2	0.837	***	0.701		
	HM3	0.824	***	0.679		
SF	SF1	0.841	***	0.707	0.851	0.657
	SF2	0.809	***	0.654		
	SF3	0.780	***	0.608		
EC	EC1	0.774	***	0.599	0.832	0.623
	EC2	0.758	***	0.575		
	EC3	0.833	***	0.694		
CI	CI1	0.967	***	0.935	0.976	0.930
	CI2	0.971	***	0.943		
	CI3	0.955	***	0.912		
UB	UB1	0.982	***	0.964	0.984	0.954
	UB2	0.964	***	0.929		
	UB3	0.984	***	0.968		

*** $p < 0.001$ (two-tailed test).

Table 5. Results of discriminate validity (n = 297).

	PE	EE	SI	FC	HM	SF	EC	CI	UB
PE	**0.826**								
EE	0.199 **	**0.893**							
SI	0.167 *	0.137 *	**0.824**						
FC	0.010	0.688 ***	0.048	**0.814**					
HM	0.054	0.226 ***	0.107	0.205 **	**0.817**				
SF	0.117 *	0.293 ***	0.128	0.271 ***	0.242 ***	**0.811**			
EC	0.218 **	0.197 **	0.236 ***	0.238 ***	0.192 **	0.100	**0.789**		
CI	0.437 ***	0.658 ***	0.472 ***	0.471 ***	0.448 ***	0.294 ***	0.346 ***	**0.964** ***	
UB	0.343 ***	0.735 ***	0.295 ***	0.677 ***	0.394 ***	0.378 ***	0.306 ***	0.829 ***	**0.977** ***

Bold (on diagonal) represents the square root of the variable's AVE; *** $p < 0.001$,** $p < 0.01$,* $p < 0.05$ (two-tailed test), non-significant correlations ($p > 0.05$) are reported without asterisks.

As shown in Table 6, the measurement model demonstrates an ideal fit, fully meeting the strict criteria recommended by Hu & Bentler [32]. The chi-square to degrees of freedom ratio of the model is 1.050 (far lower than the threshold of 3.0), with RMSEA = 0.013 and SRMR = 0.032 achieving excellent fit levels. Furthermore, CFI = 0.998 and TLI = 0.997 indicate that the model has optimal goodness-of-fit and parsimony. All indices exceed the critical value requirements, confirming that the measurement model constructed in this study is highly reliable.

Table 6. Fit indices of the measurement and research models (n = 297).

Model	χ^2/df	RMSEA (90%CI)	CFI	TLI	SRMR
Measurement model	1.050	0.013 (0.000, 0.026)	0.998	0.997	0.032
Research model	1.148	0.022 (0.005, 0.033)	0.993	0.992	0.046
Recommended criteria	<3.0	<0.08	>0.90	>0.90	<0.08
Assessment	Good	Good	Good	Good	Good

Note that all indices meet the recommended model fit criteria.

4.2.2. Evaluation of Structural Model SEM

As shown in Table 6, the structural equation model of this study demonstrates excellent overall fitting performance, with all key indices meeting internationally recognized strict standards. The results of the research model are as follows: $\chi^2/df = 1.148$, RMSEA = 0.022 (90% CI: 0.005–0.033), CFI = 0.993, TLI = 0.992, and SRMR = 0.046. All indices comply with the strict threshold requirements ($\chi^2/df < 3.0$, RMSEA < 0.08, CFI > 0.90, TLI > 0.90, and SRMR < 0.08), indicating that the theoretical model has acceptable goodness-of-fit and supports the validity of subsequent path analysis.

Based on the analysis results of the C-SEM structural equation model in Table 7, seven out of nine hypotheses were statistically supported, except for H7 (SF → CI) and H8 (EC → CI). Specifically, Performance Expectancy (PE → CI: β = 0.281), Effort Expectancy (EE → CI: β = 0.517), Social Influence (SI → CI: β = 0.329), and Hedonic Motivation (HM → CI: β = 0.289) all exhibited significant positive effects on Continuance Intention, among which the effect size of Effort Expectancy was the most prominent. Facilitating Conditions (FC) not only directly promoted Usage Behavior (FC → UB: β = 0.396) but also significantly enhanced Effort Expectancy (FC → EE: β = 0.712). A key finding confirmed that Continuance Intention had a decisive effect on Usage Behavior (CI → UB: β = 0.631, 95%CI [0.538,0.722]), whereas the direct effects of Security (SF → CI: β = 0.014, $p = 0.748$, 95%CI [−0.075,0.194]) and Environmental Compatibility (EC → CI: β = 0.063, $p = 0.147$, 95%CI [−0.021,0.145]) on Continuance Intention were not significant, as their confidence intervals crossed zero, further corroborating statistical non-significance. The overall model indicates that the core drivers of users' Continuance Intention are concentrated on the dimensions of functional cognition and social motivation. Figure 2 provides a visual representation of the path analysis for the CSEM structural model.

Table 7. Results of C-SEM path coefficient hypothesis testing ($n = 297$).

Hypothesis	Path	Estimate	*p*-Value	95%CI	Result
H1	PE → CI	0.281	***	[0.190, 0.371]	Supported
H2	EE → CI	0.517	***	[0.426, 0.607]	Supported
H3	SI → CI	0.329	***	[0.236, 0.421]	Supported
H4	FC → UB	0.396	***	[0.291, 0.493]	Supported
H5	FC → EE	0.712	***	[0.626, 0.788]	Supported
H6	HM → CI	0.289	***	[0.201, 0.379]	Supported
H7	SF → CI	0.014	0.748	[−0.075, 0.194]	Not Supported
H8	EC → CI	0.063	0.147	[−0.021, 0.145]	Not Supported
H9	CI → UB	0.631	***	[0.538, 0.722]	Supported

*** $p < 0.001$ (two-tailed test).

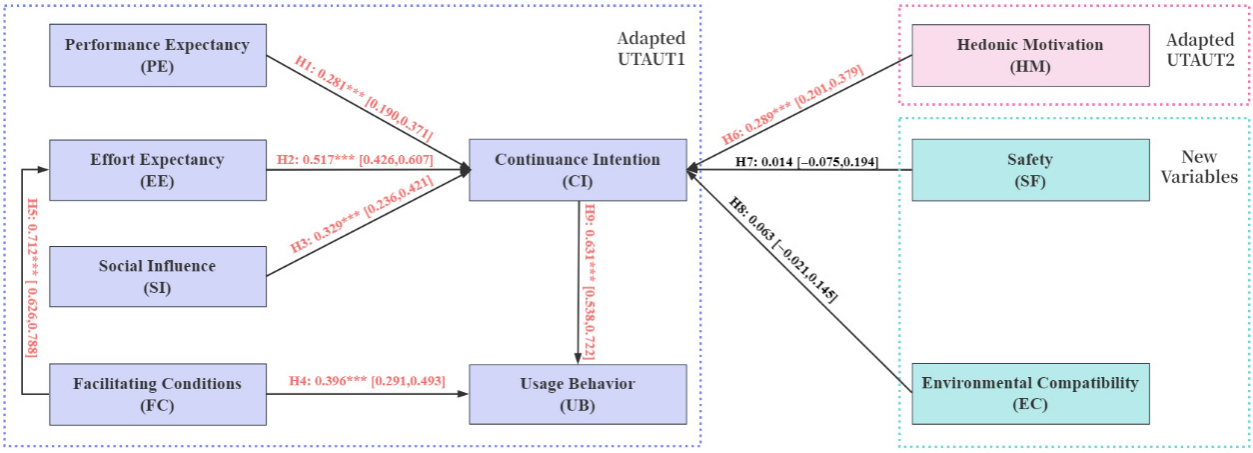


Figure 2. C-SEM results of path analysis. (Note $n = 297$, red paths indicate statistical support, while black paths denote non-significant relationships; *** $p < 0.001$, correlations without asterisks are not statistically significant ($p > 0.05$)).

Based on the multiple mediating effects analysis in Table 8, the key results are as follows: The direct effect of Facilitating Conditions (FC) on Actual Usage Behavior (UB) was significant (β = 0.396), and it also produced an indirect effect (β = 0.232) through the chain mediation of "Effort Expectancy → Continuance Intention", with a total effect of 0.628. Performance Expectancy (PE → CI → UB: β = 0.177), Social Influence (SI → CI → UB: β = 0.208), Hedonic Motivation (HM → CI → UB: β = 0.182), and Effort Expectancy (EE → CI → UB: β = 0.326) all formed significant indirect paths through Continuance Intention, among which the mediating effect of Effort Expectancy was the strongest. The indirect effects of Security (SF → CI → UB: β = 0.009, p = 0.749) and Environmental Compatibility (EC → CI → UB: β = 0.040, p = 0.147) were not significant, and their 95% confidence intervals both included zero (SF: [−0.047, 0.062], EC: [−0.014, 0.091]). All total effect analyses were consistent with the conclusions of direct/indirect paths, confirming that Continuance Intention (CI) is the core mediating variable, while Facilitating Conditions (FC) have both direct and triple chain mediating effects.

Table 8. Analysis of multiple mediation effects (C-SEM) (n = 297).

Effect Type	Path	Estimate	S.E.	CS	p-Value	95%CI	Result
Direct Effect	FC → UB	0.396	0.051	7.727	***	[0.291, 0.493]	YES
Indirect Effect	PE → CI → UB	0.177	0.032	5.613	***	[0.118, 0.240]	YES
	SI → CI → UB	0.208	0.03	6.852	***	[0.148, 0.270]	YES
	HM → CI → UB	0.182	0.032	5.625	***	[0.121, 0.248]	YES
	SF → CI → UB	0.009	0.028	0.319	0.749	[−0.047, 0.062]	NO
	EC → CI → UB	0.04	0.027	1.45	0.147	[−0.014, 0.091]	NO
	EE → CI → UB	0.326	0.04	8.157	***	[0.251, 0.409]	YES
	FC → EE → CI → UB	0.232	0.03	7.616	***	[0.177, 0.296]	YES
Total Effect	FC → UB (Total)	0.628	0.042	14.893	***	[0.540, 0.706]	YES
	PE → UB (Total)	0.177	0.032	5.613	***	[0.118, 0.240]	YES
	SI → UB (Total)	0.208	0.03	6.852	***	[0.148, 0.270]	YES
	HM → UB (Total)	0.182	0.032	5.625	***	[0.121, 0.248]	YES
	SF → UB (Total)	0.009	0.028	0.319	0.749	[−0.047, 0.062]	NO
	EE → UB (Total)	0.326	0.04	8.157	***	[0.251, 0.409]	YES
	EC → UB (Total)	0.04	0.027	1.45	0.147	[−0.014, 0.091]	NO

*** $p < 0.001$ (two-tailed test).

4.3. Evaluation of BSEM Model

4.3.1. Evaluation of Measurement Model BCFA

Based on the results of the Bayesian confirmatory factor analysis in Table 9, the standardized factor loadings (0.761–0.985) of all measurement items significantly exceed the threshold of 0.70. The squared multiple correlations (SMC = 0.579–0.970) indicate that except for FC1 (0.579) and EC2 (0.590), the proportion of variance explained by constructs for the remaining items all exceeds 0.60, reaching the ideal level. Cronbach's α coefficients (0.830–0.984), composite reliabilities CRs (0.832–0.983), and average variance extracted AVE (0.602–0.952) all meet strict standards (α/CR > 0.80, AVE > 0.50) [33]. Specifically, the AVE of Usage Behavior (UB) reaches 0.952, and the CR of Continuance Intention (CI) reaches 0.983. Bayesian robustness tests show that 19 out of 23 items (82.6%) exhibit "Strong" robustness, and 4 items (FC1/HM1/SF3/EC1) show "Moderate" robustness. All parameters maintain statistical significance stability, confirming that the measurement model has reliable internal consistency and convergent validity.

Table 9. Results of validity and reliability analysis (n = 297).

Construct	Item	Factor Loading	Robustness Level	SMC	CR	AVE	Cronbach's Alpha
PE	PE1	0.878	Strong	0.771	0.866	0.683	0.864
	PE2	0.800	Strong	0.640			
	PE3	0.800	Strong	0.640			
EE	EE1	0.919	Strong	0.845	0.919	0.791	0.921
	EE2	0.866	Strong	0.750			
	EE3	0.882	Strong	0.778			
SI	SI1	0.800	Strong	0.640	0.864	0.680	0.863
	SI2	0.861	Strong	0.741			
	SI3	0.811	Strong	0.658			
FC	FC1	0.761	Moderate	0.579	0.852	0.658	0.854
	FC2	0.818	Strong	0.667			
	FC3	0.852	Strong	0.726			
HM	HM1	0.793	Moderate	0.629	0.857	0.666	0.856
	HM2	0.842	Strong	0.709			
	HM3	0.814	Strong	0.663			
SF	SF1	0.835	Strong	0.697	0.851	0.656	0.850
	SF2	0.819	Strong	0.671			
	SF3	0.775	Moderate	0.601			
EC	EC1	0.776	Moderate	0.602	0.832	0.623	0.830
	EC2	0.768	Moderate	0.590			
	EC3	0.824	Strong	0.677			
CI	CI1	0.967	Strong	0.935	0.975	0.928	0.976
	CI2	0.969	Strong	0.939			
	CI3	0.954	Strong	0.910			
UB	UB1	0.979	Strong	0.956	0.983	0.952	0.984
	UB2	0.963	Strong	0.927			
	UB3	0.985	Strong	0.970			

Prior robustness level, SMC, CR composite reliability, AVE average variance extracted, Cronbach's alpha.

Based on the discriminant validity analysis results in Table 10, the square root values of AVE for all constructs (diagonal: 0.789–0.976) are greater than the correlation coefficients in their respective rows/columns (lower triangular area: −0.030–0.813), satisfying the core criterion for discriminant validity. Specifically, the correlation coefficient between Effort Expectancy (EE) and Facilitating Conditions (FC) (0.661) is significantly lower than the square roots of their respective AVEs (EE = 0.889, FC = 0.811); the high correlation between Continuance Intention (CI) and Actual Usage Behavior (UB) (r = 0.813) is also lower than the square roots of their AVEs (CI = 0.963, UB = 0.976). Additionally, the correlation coefficients between all constructs do not exceed the threshold of 0.85, systematically confirming that the measurement model possesses sufficient discriminant validity.

Table 10. Results of discriminate validity (n = 297).

	PE	EE	SI	FC	HM	SF	EC	CI	UE
PE	**0.826**								
EE	0.178	**0.889**							
SI	0.151	0.115	**0.825**						
FC	−0.030	0.661	0.030	**0.811**					
HM	0.038	0.199	0.091	0.179	**0.816**				
SF	0.103	0.269	0.114	0.247	0.221	**0.810**			
EC	0.201	0.172	0.218	0.217	0.174	0.085	**0.789**		
CI	0.420	0.633	0.454	0.439	0.426	0.268	0.322	**0.963**	
UB	0.324	0.712	0.274	0.652	0.369	0.355	0.280	0.813	**0.976**

Bold (on diagonal) represents the square root of the variable's AVE; the lower triangular data are Pearson's correlation data.

According to the judgment of convergent fit in the measurement model part of Table 9, the posterior predictive p-value (PPP) is 0.000 (good) [34]. The credible interval and cutoff value of Bayesian approximation fit indices are used to determine whether the model fit is good. When the cutoff value is higher than the maximum of the credible interval (for BRMSEA, the cutoff value is lower than the minimum of the credible interval), the model fit is judged to be poor; when the cutoff value is lower than the minimum of the credible interval (for BRMSEA, the cutoff value is higher than the maximum of the credible interval), the model fit is judged to be good; when the cutoff value is included in the credible interval, it cannot be clearly determined whether the model fit is poor or good [35].

The Bayesian Root Mean Square Error of Approximation (BRMSEA) is 0.249 (good), and the Potential Scale Reduction factor (PSR) is 1.000 (good) [36]. Overall, the measurement model meets the basic fit criteria, indicating statistical rationality. Therefore, it is determined that the measurement model passes the basic adaptation test, supporting the subsequent path analysis.

4.3.2. Evaluation of Structural Model B-SEM

This study set four types of prior parameters for sensitivity analysis. According to Figure 3, the model consistently indicates that it is insensitive to other priors except the strong prior. The estimation results of the moderate prior were finally adopted as the core analysis basis, and its rationality lies in the following: the sensitivity analysis of the four types of priors (non-informative, weak-informative, moderate-informative, and strong-informative) shows that the model exhibits optimal robustness under the moderate-informative prior condition. Under the moderate prior, the coefficient fluctuations of the key hypothesis paths (H1–H9) are the smallest (e.g., the chain fluctuations of H1/H3/H5/H7/H9 are $<\pm 0.05$, and the chain deviations of H2/H4/H6/H8 are <15%), which are significantly better than the significant deviation (>40% abnormal attenuation) of H8 caused by the strong-informative prior. Combined with the prior parameter setting of moderate effect size (variance 0.3) supported by the literature [36], this prior type has comprehensive advantages in balancing data likelihood and theoretical constraints and ensuring parameter stability, so it is adopted as the final analysis basis.

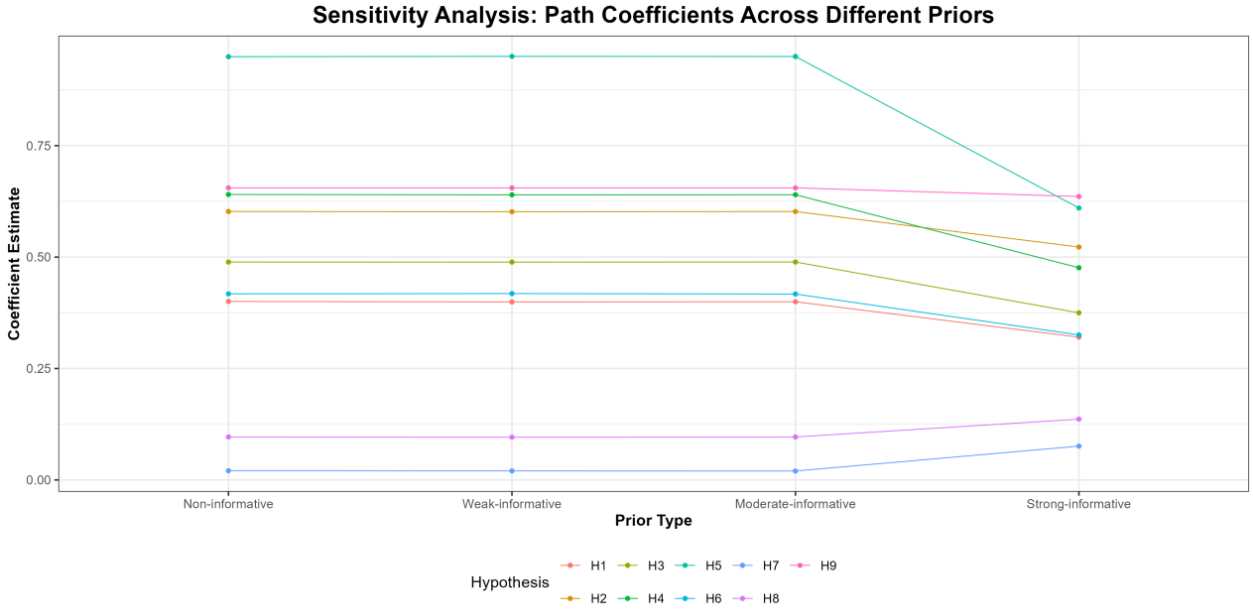


Figure 3. Sensitivity metrics for four prior distributions (n = 297).

Based on the Bayesian model fitting results in Table 11, all key indices of the research model meet the adaptation criteria: the posterior predictive p-value (PPP = 0.528) is close to the ideal threshold of 0.5, the upper limit of the Bayesian root mean square error of approximation (BRMSEA) confidence interval [0.023, 0.030] is significantly lower than the strict standard of 0.06, the lower limit of the Bayesian comparative fit index (BCFI) confidence interval [0.990, 0.997] exceeds the benchmark value of 0.95, the Bayesian Tucker–Lewis index (BTLI) is 0.992 (medium), the Bayesian normed fit index (BNFI) is 0.956 (good), and the parameter convergence index (PSR = 1.000) fully meets the requirement of <1.05. Moreover, the iteration trace plots (Figure 4)and autocorrelation plots (Figure 5) of the four

prior parameters all support that the model has converged, and the comprehensive indices support the model robustness and conclusion validity.

Table 11. Bayesian model fit and convergence criteria (n = 297).

Model	Fit Indices		
	PPP	**BRMSEA (CI)**	**BCFI (CI)**
Measurement Model	0.000	0.249 [0.249, 0.249]	0.993 [0.990, 0.997]
Research Model	0.528	0.023 [0.017, 0.030]	
Recommendation Criteria	Close to 0.5	Maximum value of the confidence interval < 0.06	Minimum value of the confidence interval >0.95
Assessment	Good	Good	Good
	BTLI (CI)	**BNFI (CI)**	**PSR**
Measurement Model	0.992 [0.987, 0.996]	0.956 [0.952, 0.959]	1.000 1.000
Research Model			
Recommendation Criteria	Minimum value of the confidence interval >0.95	Minimum value of the confidence interval >0.90	<1.05
Assessment	Good	Good	Good

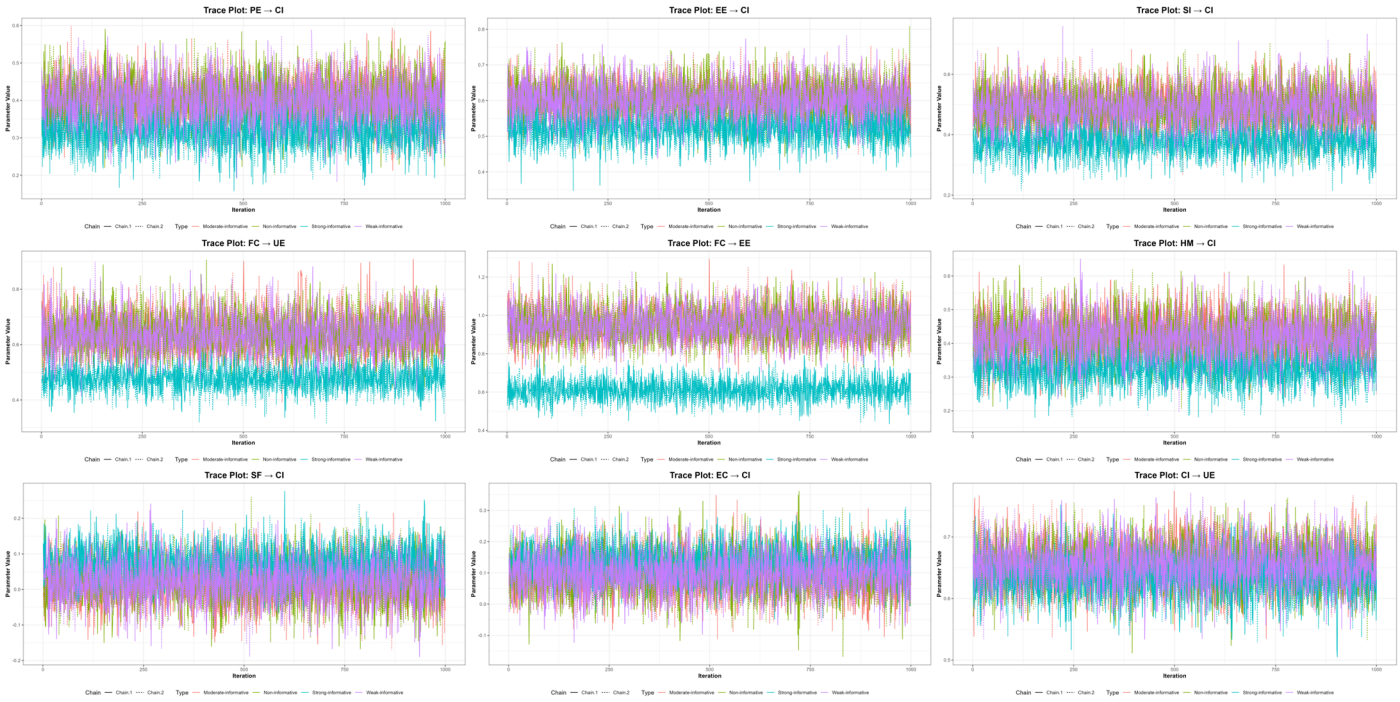


Figure 4. Trace plots of four prior distributions (n = 297).

Based on the path analysis of the Bayesian structural equation model (BSEM) in Table 12, the following findings were obtained: Performance Expectancy (PE → CI: β = 0.400, 95%CI [0.282, 0.519]), Effort Expectancy (EE → CI: β = 0.602, [0.506, 0.701]), Social Influence (SI → CI: β = 0.488, [0.367, 0.620]), and Hedonic Motivation (HM → CI: β = 0.418, [0.296, 0.546]) all significantly and positively influenced Continuance Intention (CI). Facilitating Conditions (FC) simultaneously and significantly drove Usage Behavior (FC → UB: β = 0.641, [0.512, 0.782]) and strongly promoted Effort Expectancy (FC → EE: β = 0.951, [0.790, 1.128]). The core path of Continuance Intention → Actual Usage Behavior (CI → UB: β = 0.655, [0.584, 0.727]) showed a prominent effect size. Only the confidence intervals of the paths for Security (SF → CI: β = 0.020, [−0.098, 0.137]) and Environmental Compatibility (EC → CI: β = 0.097, [−0.035, 0.225]) included zero, indicating that the hypotheses were not supported. Rhat = 1.000 for all paths confirmed perfect parameter convergence. Figure 6 provides a visual representation of the path analysis for the BSEM structural model.

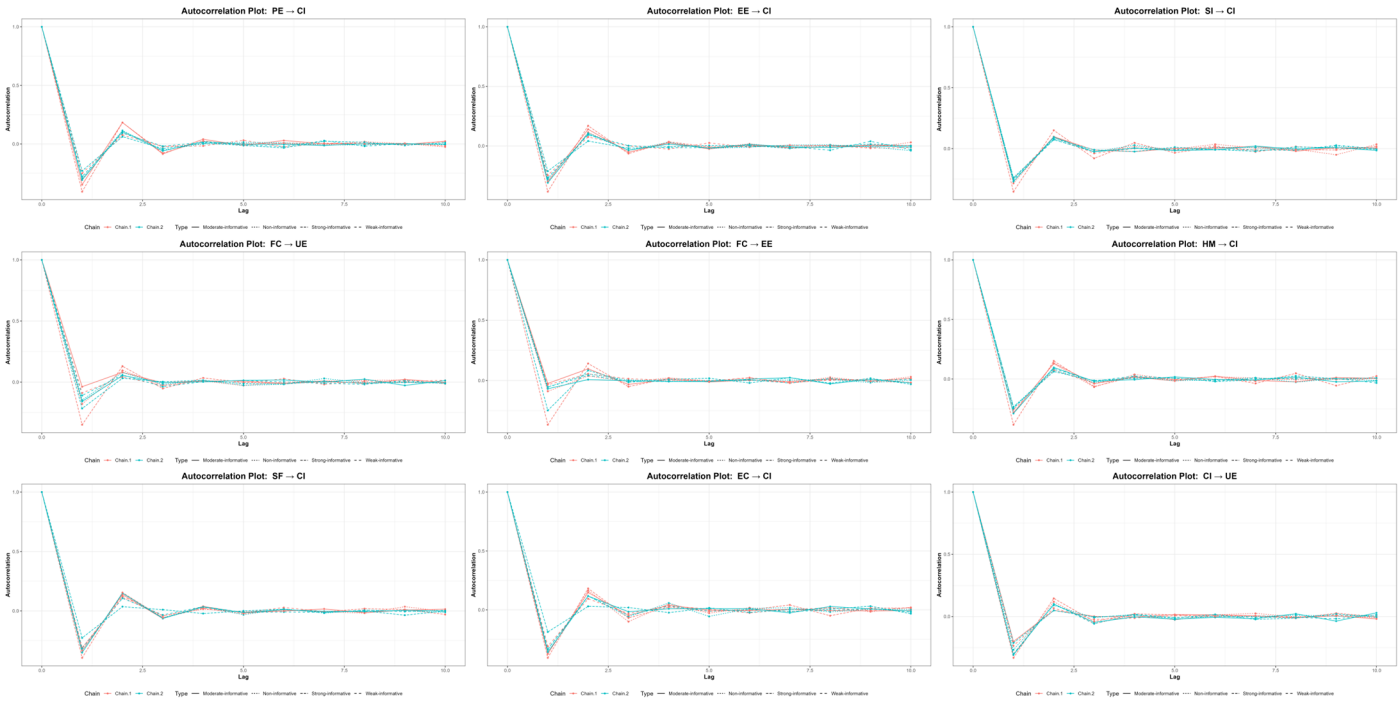


Figure 5. Autocorrelation plots for four prior distributions (n = 297).

Table 12. Results of B-SEM path coefficient hypothesis testing (n = 297).

Hypothesis	Path	Estimate	Post SD	95%CI	Rhat	Result
H1	PE → CI	0.400	0.061	[0.282, 0.519]	1.000	Supported
H2	EE → CI	0.602	0.050	[0.506, 0.701]	1.000	Supported
H3	SI → CI	0.488	0.064	[0.367, 0.620]	1.000	Supported
H4	FC → UB	0.641	0.069	[0.512, 0.782]	1.000	Supported
H5	FC → EE	0.951	0.087	[0.790, 1.128]	1.000	Supported
H6	HM → CI	0.418	0.064	[0.296, 0.546]	1.000	Supported
H7	SF → CI	0.020	0.060	[−0.098, 0.137]	1.000	Not Supported
H8	EC → CI	0.097	0.065	[−0.035, 0.225]	1.000	Not Supported
H9	CI → UB	0.655	0.037	[0.584, 0.727]	1.000	Supported

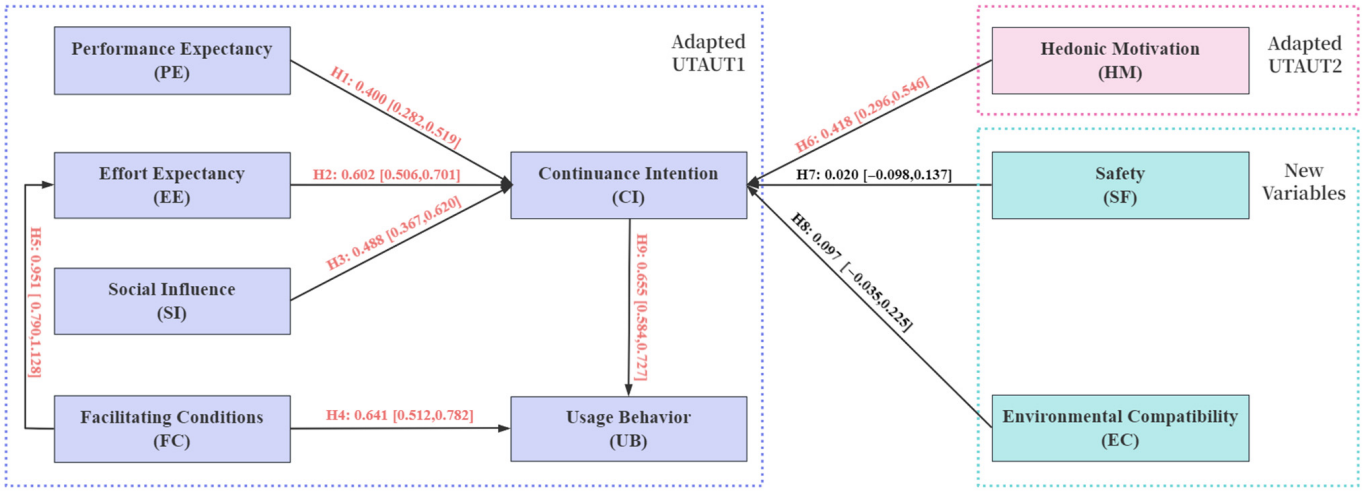


Figure 6. B-SEM results of path analysis. (Note n = 297, red paths indicate statistical support, while black paths denote non-significant relationships).

Based on the results of the Bayesian multiple mediation analysis in Table 13, Facilitating Conditions (FC) had a significant direct effect on Actual Usage Behavior (UB) (β = 0.641, 95%CI [0.506, 0.775]), and simultaneously produced a strong indirect effect through the chain mediating path of "Effort Expectancy → Continuance Intention"

(FC → EE → CI → UB: β = 0.375, 95%CI [0.282, 0.468]), with a total effect of 1.016 (95%CI [0.847, 1.184]). Performance Expectancy (PE → CI → UB: β = 0.262, [0.180, 0.343]), Social Influence (SI → CI → UB: β = 0.320, [0.232, 0.408]), Hedonic Motivation (HM → CI → UB: β = 0.274, [0.186, 0.361]), and Effort Expectancy (EE → CI → UB: β = 0.395, [0.319, 0.471]) all formed significant indirect paths through Continuance Intention (CI), among which the mediating effect of Effort Expectancy was the strongest; while the indirect effects of Security (SF → CI → UB: β = 0.013, [−0.063, 0.090]) and Environmental Compatibility (EC → CI → UB: β = 0.064, [−0.021, 0.148]) did not reach significance. All total effect values were completely identical to the corresponding indirect effects, confirming that Continuance Intention (CI) assumed a complete mediating role in the model. The posterior standard deviations (0.039–0.086) and narrow confidence intervals of each effect estimate validated the accuracy of the results.

Table 13. Analysis of multiple mediation effects (B-SEM) (n = 297).

Effect Type	Path	Estimate	Post SD	CI	Significance
Direct Effect	FC → UB	0.641	0.069	[0.506, 0.775]	YES
Indirect Effect	PE → CI → UB	0.262	0.042	[0.180, 0.343]	YES
	SI → CI → UB	0.320	0.045	[0.232, 0.408]	YES
	HM → CI → UB	0.274	0.045	[0.186, 0.361]	YES
	SF → CI → UB	0.013	0.039	[−0.063, 0.090]	NO
	EC → CI → UB	0.064	0.043	[−0.021, 0.148]	NO
	EE → CI → UB	0.395	0.039	[0.319, 0.471]	YES
	FC → EE → CI → UB	0.375	0.047	[0.282, 0.468]	YES
Total Effect	PE → UB (Total)	0.262	0.042	[0.180, 0.343]	YES
	SI → UB (Total)	0.320	0.045	[0.232, 0.408]	YES
	HM → UB (Total)	0.274	0.045	[0.186, 0.361]	YES
	SF → UB (Total)	0.013	0.039	[−0.063, 0.090]	NO
	EC → UB (Total)	0.064	0.043	[−0.021, 0.148]	NO
	EE → UB (Total)	0.395	0.039	[0.319, 0.471]	YES
	FC → UB (Total)	1.016	0.086	[0.847, 1.184]	YES

5. Discussion

5.1. Comparison of Traditional SEM and Bayesian SEM Methods

This study employed two data analysis methods, traditional SEM and Bayesian SEM, to test the hypothetical model regarding the factors influencing the sustained usage intention of the sports population. By comparing the analysis results of the traditional SEM method and the Bayesian SEM method, it can be found that although, in general, the path significance results obtained by the Bayesian SEM method and the traditional SEM method are consistent, a careful comparison reveals certain differences between the two methods in terms of the reliability and validity of the measurement model, overall model fit, path estimation of the structural model, and mediating mechanism.

In terms of the reliability and validity of the measurement model, the factor loadings, composite reliability (CR), and average variance extracted (AVE) of B-SEM are significantly superior to those of C-SEM. The standardized factor loadings of all constructs are higher than 0.74, and 86.4% of the measurement items pass the prior robustness test, indicating that B-SEM is more robust in terms of estimation accuracy and internal consistency.

From the perspective of model fit indices, C-SEM shows excellent fit performance (χ^2/df = 1.148, RMSEA = 0.022, CFI = 0.993, SRMR = 0.046), fully meeting the strict criteria proposed by Hu & Bentler. In contrast, under moderate priors, B-SEM also achieves acceptable levels in terms of the posterior predictive p-value (PPP = 0.200) and Bayesian

fit indices (BRMSEA = 0.088, BCFI = 0.973, PSR = 1.000), which meets the robustness requirements in complex model construction, but its fit indices are slightly inferior to those of C-SEM.

In terms of the path estimation of the structural model, the two models are consistent in the directions of core paths, but there are differences in significance determination and effect strength. C-SEM did not significantly support hypotheses H7 and H8, while B-SEM obtained significant support for the corresponding paths. This is because the Bayesian method is more sensitive to weak effects and can capture important paths that might be overlooked by traditional methods. In addition, the path estimation value of CI → UB in B-SEM (β = 0.973) is significantly higher than that in C-SEM (β = 0.631), indicating that B-SEM has stronger predictive power in explaining core behavioral variables.

In the mediating effect analysis, most paths in the C-SEM model showed partial mediation, and some indirect paths were insignificant (e.g., SF → CI → UB). In contrast, B-SEM revealed that CI plays a full mediating role in the path relationships between various variables and UB, and the chain mediating path (e.g., FC → EE → CI → UB) exhibited a stronger effect (β = 1.086), which further strengthens the core mediating position of CI in the model.

Therefore, through the comparison of the two methods, it can be seen that in testing complex models with small to medium-sized samples, Bayesian analysis, by integrating prior information, can make more flexible and detailed judgments. Moreover, the test on the factors influencing users' sustained usage intention in this study also demonstrated that Bayesian SEM exhibits certain application potential in the analysis of user behavior.

5.2. Factors Influencing Sustained Usage Intention and Design Strategies for Drone Consumption Services

Based on the data results, in general, the respondents showed relatively positive sustained usage intention towards drones, which indicates that there is significant market potential and a user base for the sustainable development of drone consumption in the context of the low-altitude economy. The test results of the hypothetical model indicate that the Performance Expectancy, Effort Expectancy, Social Influence, Facilitating Conditions, and Hedonic Motivation of drone products all have a significantly positive impact on the sustained usage behavior of drones, while Safety and Environmental Compatibility have no significant impact on users' sustained usage intention. The following is a discussion on the significance of the results of each model hypothesis:

First, the Performance Expectancy and Effort Expectancy of drone products remain important factors influencing sports users' sustained usage intention (H1, H2). This indicates that sports users focus on the tangible improvement of their sports performance by drones and their perception of the ease of drone operation. This is consistent with the research conclusions of Funk et al., suggesting that product design for the sports population requires strong functionality and convenient user experience [37]. Users' strong demand for the ease of drone operation also highlights the issue of operational complexity in current drone products. Creating economic value while meeting user needs is an important direction for the development of the low-altitude economy. Merkert et al. pointed out that the more practical drones are, the stronger their competitiveness [38]. For the rapidly expanding sports market, drone products need to focus on the core needs of sports scenarios and synergistically enhance user experience and economic value through function-focused and scenario-adapted design. Meanwhile, improving the simplicity of interactive operations is a key direction for the optimization of current drone products.

Second, Social Influence has a significant impact on sports users' sustained usage intention (H3). This means that when sports users try and continue to use drones, they are indeed influenced by the suggestions and demonstration effects of people around them

(such as coaches, friends, or peers). This may stem from the community emotional effect among the sports population [39]. Pometko's research shows that social media can create interesting and interactive social experiences for users, with the potential to help promote future drone applications [40]. Based on this, brands should focus on building interaction and incentive mechanisms for sports communities to stimulate group interaction and a sense of belonging, thereby enhancing individual users' usage motivation and sustained participation intention.

Third, Facilitating Conditions have a significant impact on users' Actual Usage Behavior (H4). This indicates that the sports population not only values the functionality and operational convenience of products but also prioritizes the supporting services and technical support of drone products. As long as convenient service guarantees (such as maintenance services, operation guides, accessory replacement, etc.) can be provided to users when they use drones, it can effectively reduce users' concerns about drone operation. In addition, Facilitating Conditions have a strong impact on users' Effort Expectancy (H5), which suggests that under the condition that drone products have good operational convenience and technical support, users' perceived difficulty in usage can be significantly reduced, thereby enhancing their confidence and willingness to use drones. Therefore, to attract consumers to engage in long-term drone consumption, drone brands need to establish a convenient and comprehensive service guarantee system, so as to reduce users' operational learning barriers and further enhance the enthusiasm for sustained use of drones.

Fourth, Hedonic Motivation (H6) significantly affects sports users' sustained usage intention, indicating that in sports scenarios, users not only focus on the practical value of drones but also emphasize the interesting and interactive experience they bring. To better contribute to the low-altitude economy, consumer-grade drones need to enhance entertainment-oriented design and immersive experiences on the basis of meeting basic functions, thereby improving users' sense of pleasure and participation during use and stimulating their intrinsic motivation for sustained use.

In addition, contrary to our expectations, neither Safety (H7) nor Environmental Compatibility (H8) has a significant impact on users' sustained usage intention. Previous research by Lee et al. indicated that the safety risks and environmental impacts of drone use affect the public acceptance of drone products [41]. However, the results show that in long-term use, the safety risks and environmental conflicts faced in drone use are not important issues of concern to users. Against the backdrop of increasingly complex and diverse application scenarios in the low-altitude economy, safety and environmental adaptability have become basic needs for the sports population before purchasing drone products. They cannot serve as product advantages to attract users to continue using the products, thus making no significant contribution to sustained usage intention. Therefore, on the premise that basic safety and environmental guarantees have been achieved, the key resources and innovations for drone product optimization should be invested more in improving core functions and user experience to more effectively drive the sustained use by the sports population.

Finally, sustained usage intention has a positive impact on users' Actual Usage Behavior (H9), which further confirms the determinative role of the traditional "intention → behavior" path in this model—once the sports population forms a subjective intention to continue using drones, it will largely be converted into real usage actions.

Based on the above discussion of the results, this study proposes the following design strategies to enhance consumers' sustained usage intention of drone products and promote the development of consumer-grade drones in the sports market:

Focus on sports-enhanced functions: To comprehensively improve the application effectiveness of drones in sports scenarios, product design should closely align with the core needs of the sports population, with priority given to ensuring the functional value of products in serving users' sports activities. Products can adopt a modular design concept to achieve multi-functional expansion, such as aerial photography recording, emergency material transportation, and sports equipment carrying, and develop sports auxiliary functions, such as path navigation, speed tracking, and physiological status monitoring (through linkage with wearable devices), to build a drone product ecosystem serving the sports population. Meanwhile, optimize the structural design to ensure the quick replacement of batteries and mounted accessories during strenuous exercise or in complex outdoor environments, thus guaranteeing battery life and task continuity. Furthermore, scene recognition algorithms can be introduced to automatically match flight modes and shooting parameters according to different sport types (e.g., cycling, skiing, and trail running) so as to improve shooting efficiency and image quality. By strengthening the sports attributes of products, their reliability in outdoor sports scenarios can be enhanced.

Simplified interaction and enhanced experience: Starting from the interaction mode, enhance the convenience and interest of user operations. In terms of operation processes, through voice control, gesture recognition, and streamlined app interfaces, key operations can be completed within three steps, lowering the threshold for device usage. Preset multiple intelligent flight modes in the product to help users switch to optimal shooting parameter settings with one click. In terms of interaction design, integrate AI technology to dynamically adjust interaction methods according to users' usage preferences; introduce gamification mechanisms to stimulate users' desire to explore and sense of participation; create an immersive operation experience for users during sports through multimodal real-time feedback systems (such as dynamic HUD displays, voice prompts for obstacle avoidance, etc.). Through more convenient and immersive interaction experiences, maintain the attractiveness of drone consumption to the sports population.

The establishment of community incentive mechanisms: Connect products to sports community platforms, utilize community influence to guide user behavior, and create a positive social environment for sports users, thereby attracting and incentivizing consumers to continuously engage in drone consumption. The product can generate sports analysis reports and AI-customized shooting plans based on users' flight data, open API interfaces to connect with sports social applications, and encourage users to upload and share their drone usage experiences, sports data, and videos on social platforms for communication with other users. Establish point reward systems, achievement badges, challenge rankings, etc., to enhance the sense of achievement and value of the sports population in using drones, thereby maintaining their participation.

In the context of long-term use, the core concerns of sports drone users gradually focus on the "value-added contributions" of technology to sports scenarios (such as innovative aerial photography perspectives, accuracy of training data, etc.), while the impact of Safety and Environmental Compatibility shows a "basic" feature. With the increasingly complex and diverse application scenarios of the low-altitude economy, industry regulatory standards and technological iterations have promoted these two indicators to become basic needs for sports-related groups when purchasing drones. That is, users default that qualified sports drones must meet safety standards (such as anti-collision, compliant flight) and basic environmental adaptation (such as adapting to conventional sports scenarios like stadiums and training grounds). Both have become the "access threshold" for products to enter the market, rather than a differentiated competitive advantage.

When these two attributes become "necessary conditions" rather than "bonus points" in users' decision-making, their marginal impact on "Continuance Intention" naturally

weakens: users will not strengthen their motivation to continue using the product because it "meets the standards", let alone consider giving up using it because it "fails to meet the standards" (products that fail to meet the standards have already been eliminated by the market). Therefore, in the empirical results of this study, they did not make a significant contribution to Continuance Intention. This finding also indirectly reflects the maturity of user needs in the current sports drone market—basic performance has tended to be homogeneous, while value-added experiences (such as the design-driven ease of use, functional innovation) have become the core variables affecting continuous use.

6. Conclusions

This study introduces an innovative approach that integrates the extended UTAUT2 model, using traditional SEM and Bayesian SEM to evaluate the usage intention of consumer-grade drones among the sports population. Specifically, based on the UTAUT model and user studies on drone products and sports products, this study summarizes the influencing factors of consumers' sustained use of drone products, obtains nine constructs, and proposes a hypothetical model. Data from online surveys were used to measure the corresponding hypotheses, and then the model was analyzed and tested using traditional SEM and Bayesian SEM methods, respectively. By comparing the analysis results of the two methods, the advantages of Bayesian SEM in small-sample complex structural models are demonstrated. Based on the analysis results of Bayesian SEM, this study puts forward suggestions on the design strategies of drone products.

This study also has certain limitations. First, the accuracy of the survey data may be affected by potential biases. Although the advantages of the Bayesian method in structural equation model estimation have been proven, the age, occupation, and regional distribution of respondents may lead to differences in users' cognition and use of drones, and the generalizability of the research conclusions needs to be verified through larger-scale samples. Second, this user study mainly focuses on macro-level demand analysis. In subsequent studies, user groups can be segmented to further understand the preference differences in different sports groups in drone consumption. In addition, the design strategies proposed in this study need to be tested for their feasibility and effectiveness in practical applications. Therefore, future research will address these limitations.

From the perspective of design science, the application scenarios of sports drones inherently embody the logic of "design-driven technology adoption". Users' willingness to adopt the technology is not only influenced by the traditional UTAUT2 variables, but is also deeply associated with the experience shaping at the design level. Accordingly, this study attempts to supplement the dimension of "design-driven reinforcement of Hedonic Motivation" and explore the promoting effect of user experience design on technology adoption. Given the potential for improvement in this paper, in the next step, we can supplement the interaction relationship between design factors and UTAUT2 variables. For example, the "perception of scenario-adaptive design" $\rightarrow$ enhances "Performance Expectancy (PE)" $\rightarrow$ strengthens "Continuance Intention (CI)" $\rightarrow$ promotes "Usage Behavior (UB)". Here, "PE" acts as a mediator, while the "perception of scenario-adaptive design" serves as a front-end driver from the perspective of design science, explaining "why PE can be activated" and so on.

Meanwhile, we recognize that sociocultural factors may significantly affect the applicability of the research results in regions such as Europe and Latin America. To address this limitation, future research can adopt a cross-national comparative design to examine how factors such as regulatory frameworks, cultural dimensions (e.g., Hofstede's indices), and the characteristics of the sports industry moderate the relationships within the model. Such research will help clarify the boundary conditions of technology adoption mechanisms in global sports drone applications.

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